RetailFlow POS

Complete Setup, Deployment & Operations Guide

Native Windows Desktop (EXE), Web Terminal, Customer Cloudflare Architecture & Central D1 Sync

Version 1.0.0 / 1.1.0 | Production Release | Cloudflare D1 Enabled

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Key Enterprise Capabilities Included:

- Sub-50ms local barcode scanning and bill checkout with 100% offline functionality.
- Real-time multi-counter inventory stock decrement & customer Khata credit balance sync.
- Dynamic NPCI UPI QR code generation on checkout screens, thermal receipts, and A4 invoices.
- Zero-terminal 1-click Cloudflare onboarding wizard for private D1 database & R2 backups.

1.0 Architecture Overview

RetailFlow POS employs an Offline-First Hybrid Architecture. Cash counters run locally on Windows Desktop (Tauri) or Web browsers (Tablets/Mobiles) with zero internet dependency. When online, registers replicate transactions asynchronously to a central Cloudflare D1 distributed SQLite database.

2.0 Installation & Running Options

Option A: Windows Setup Installer (.exe)

RECOMMENDED

`publish/RetailFlow_POS_v1.0.0_Setup.exe (3.54 MB)`

Standard 1-click Windows installer. Installs the application to Program Files, creates Start Menu and Desktop shortcuts, registers the uninstaller, and configures native auto-updating.

Option B: Enterprise MSI Package (.msi)

`publish/RetailFlow_POS_v1.0.0_Enterprise.msi (4.77 MB)`

Windows Installer package built for IT administrators. Supports silent network deployment via Active Directory Group Policy (GPO), Microsoft Intune, and remote management tools.

Option C: Standalone Portable Executable (.exe)

`publish/RetailFlow_POS_Portable.exe (9.93 MB)`

Zero-installation portable binary. Can be executed directly from a USB drive or local directory. Ideal for testing or running on secondary billing counters immediately.

Option D: Web Application & Local Network / Tablet POS

`http://localhost:3000 or http://<local-ip>:3000`

Progressive Web Application (PWA). Run on iPads, Android tablets, or smartphones on the store Wi-Fi network without installing software. Scans barcodes using device camera.

Hardware & Software Requirements:

- Windows: Windows 10 or Windows 11 (64-bit). Microsoft Edge WebView2 runtime (pre-installed on Win 10/11).
- Hardware: 4 GB RAM minimum (8 GB recommended), Intel Core i3 or equivalent, 200 MB disk space.
- Peripherals: Compatible with all ESC/POS 80mm/58mm thermal receipt printers, USB/Bluetooth barcode scanners.

3.0 First-Time Store Setup Wizard

On initial application launch, RetailFlow POS greets the store manager with a 4-step Store Setup Wizard. Alternatively, settings can be updated anytime under Settings > Shop Details & Logo.

Setting Field	Description & Format	Used In
Shop Name	Legal business or trade name displayed on bills.	All Invoices & Header
Store Logo	PNG/JPEG/SVG logo. Embedded in base64.	A4 & Thermal Bills
GSTIN / Tax ID	15-digit Indian GST number (e.g. 32ABCDE1234F1Z5).	Tax Invoices & GST Reports
Currency Symbol	Default ₹ (INR), $, €, or any custom unicode symbol.	POS UI & Receipts
Merchant UPI ID	Valid NPCI VPA (e.g. merchant@okaxis or mobile@paytm).	Dynamic QR Payments
Place of Supply	State code and name (e.g. 32-Kerala).	CGST / SGST Invoices

3.1 Dynamic NPCI UPI QR Payment Engine

RetailFlow POS includes a built-in NPCI-compliant Dynamic UPI QR Engine. When the cashier tenders an order as UPI, the system calculates the exact net payable amount and generates a dynamic QR code containing the transaction note, amount, and store VPA:

```
upi://pay?pa=merchant@okaxis&pn=RetailFlow+Mart&am=450.00&cu=INR&tn=Invoice+INV-2026-1042
```

Where Dynamic QR appears automatically:

1. Cashier Checkout Screen: Full-screen modal with amount auto-filled for customer phone scan.
2. Thermal Receipts (80mm/58mm): ESC/POS raster QR printed at bottom for instant customer record.
3. A4 GST Tax Invoices: High-resolution vector QR with B2B/B2C invoice details and payment link.

High-Contrast Typography Engine (Zero Invisible Typing Bugs)

All input fields, search bars, customer select boxes, and payment dialogs feature explicit high-contrast foreground rules and vibrant caret-color indicators

4.0 Customer Self-Service Cloudflare Setup Wizard

Store owners are never forced to use complex CLI tools like Wrangler or terminal commands. RetailFlow POS includes an interactive, 4-step visual onboarding assistant directly in the Settings menu.

Step 1: Get Free Cloudflare Credentials

1. Log into your free Cloudflare account at cloudflare.com.
2. Navigate to Workers & Pages to copy your 32-character Account ID.
3. Click My Profile > API Tokens > Create Token.
4. Select the "Edit Cloudflare Workers" template (grants access to Workers, D1, R2, KV).
5. Click "Continue to Summary" > "Create Token" and copy the token string.

Step 2: Enter & Verify in POS Settings

1. Open RetailFlow POS > Settings > Cloudflare Sync & Backup.
2. Click "Launch Setup Wizard".
3. Paste your Cloudflare Account ID and API Token.
4. Click "Verify & Continue". The app validates permissions directly against Cloudflare v4 REST API.

Step 3: 1-Click Auto-Provisioning Engine

1. Click "Auto-Provision Cloud Infrastructure".
2. POS creates central D1 distributed SQLite database: "retailflow_pos_db".
3. Automatically executes database schema migrations (sales, products, customers, mutation journal).
4. Creates Cloudflare R2 backup storage bucket: "retailflow-backups".
5. Configures serverless Cloudflare Worker API gateway.

Step 4: Diagnostics & Live Dashboard Access

1. The wizard tests end-to-end connectivity with animated green checkmarks.
2. Click "Save & Connect POS Terminals".
3. Click "Open Dashboard" to launch your live Executive Cloud Dashboard Web Page!

Live Cloud Executive Dashboard Web Page (GET /)

Once provisioned, your store URL (e.g. https://retailflow-pos-cloud.<subdomain>.workers.dev) hosts a live executive dashboard accessible from any pho

5.0 Centralised Database (Cloudflare D1) & Sync Engine

RetailFlow POS transitions from siloed local registers into a Globally Distributed Central Database powered by Cloudflare D1 (Distributed SQLite). All counters (Counter 1 Desktop EXE, Counter 2 Web, Mobile tablets) share a single source of truth.

1. Multi-Counter Real-Time Stock Decrement

When Counter 1 (Windows EXE) sells 3 units of an item, Cloudflare D1 executes an atomic decrement in the products table. Within seconds, Counter 2 (Web Tablet) pulls the mutation and decrements its local stock count, eliminating double-selling.

2. Consolidated Customer Khata (Store Credit Ledger)

Customer credit limits, current outstanding balances, and loyalty points synchronize across all terminals. If a customer makes a partial repayment at Counter 1, their updated credit balance is immediately honored at Counter 2.

3. Offline-First Resilience (Zero Latency at Checkout)

Cashiers never wait for network requests during checkout. Scanning barcodes and completing transactions takes under 50ms locally. Mutations queue in local storage and push asynchronously when connectivity is present.

4. Central Mutation Replication Journal (OpLog)

All inserts, updates, and stock adjustments append to mutation_journal with an auto-incrementing sequence_id. Terminals pull only incremental changes since their last sync sequence, minimizing bandwidth consumption.

Central Relational Schema Tables in Cloudflare D1:

Table Name	Enterprise Purpose & Multi-Terminal Functionality
`stores`	Store branding, GSTIN, UPI ID, logo base64, tax preferences.
`products`	Master catalog, SKU, barcode, purchase & selling price, GST slab, stock quantity.
`customers`	Customer master, tier, credit balance, credit limit, total spent, loyalty points.
`sales & items`	Consolidated transactions, items sold, tenders (Cash/UPI/Card/Credit), cashier.
`mutation_journal`	Auto-incrementing delta log powering multi-terminal sync without data loss.

6.0 Centralised Auto-Updates Pipeline

When code changes are made, updates distribute seamlessly to all terminals:

Windows Desktop (EXE) Auto-Updater

- Powered by native Tauri v1 Auto-Updater engine.
- Cryptographically verifies Ed25519 signatures.
- In-app update modal with changelog and progress bar.
- 1-Click "Download & Restart" applies differential patch.
- Local SQLite and settings preserved with zero data loss.

Web Application Continuous Deployment

- Zero-downtime continuous deployment on Cloudflare Pages.
- Background client polling checks /version.json every 5m.
- Unobtrusive floating banner notifies cashier on new version.
- 1-Click "Refresh & Apply" clears cache and updates PWA.
- Never interrupts active transactions or tender modals.

6.1 Dual Day-End Backups & Disaster Recovery

Dual-Layer Redundant Backup Architecture:

1. Local Installation Directory: Whenever a cashier closes a shift or triggers a manual backup, a JSON snapshot saves directly into the installation /backu

2. Cloudflare R2 Encrypted Storage: Simultaneously packages encrypted snapshots to Cloudflare R2 for offsite catastrophe protection.

3. 1-Click Restore: Browse remote cloud backups in POS Settings and restore point-in-time state with 1 click.

GitHub Release Publishing Commands (publish/ directory):

```
git tag v1.0.0
git push origin v1.0.0
gh release create v1.0.0 publish/*.exe publish/*.msi publish/*.zip --title "v1.0.0"
```

GitHub Actions automatically signs Windows binaries and generates latest.json updater manifest.